

St George Girls High School

Year 12

Mid-HSC Course Examination

2010



Mathematics

Extension 1

General Instructions

- Working time – 1½ hours
- Reading time – 5 minutes
- Write using blue or black pen.
- Board-approved calculators may be used.
- A table of standard integrals is provided.
- All necessary working should be shown in every question.
- Write on one side of the page only.
- Start each question in a **new booklet**.

Total marks – 72

- Attempt Questions 1 – 6
- All questions are of equal value

Question 1 – (12 marks) – (Start a new booklet)

Marks

- a) Find the value of k

2

$$\log_e 5 - \frac{1}{2} \log_e 16 = \log_e k$$

- b) (i) Express $\frac{2\pi}{9}$ radians in degrees.

1

- (ii) Find the exact value of

2

$$\cos\left(\frac{-11\pi}{6}\right) + \sin\left(\frac{2\pi}{3}\right)$$

- c) Find the derivative of $y = \log_e \left(\frac{x+1}{\sqrt{x}}\right)$ expressing your answer as a single algebraic fraction.

3

- d) Find the equation of the normal to the curve $y = \log_e(x^2 + 2)$ at the point where $x = 1$

4

Question 2 – (12 marks) – (Start a new booklet)

Marks

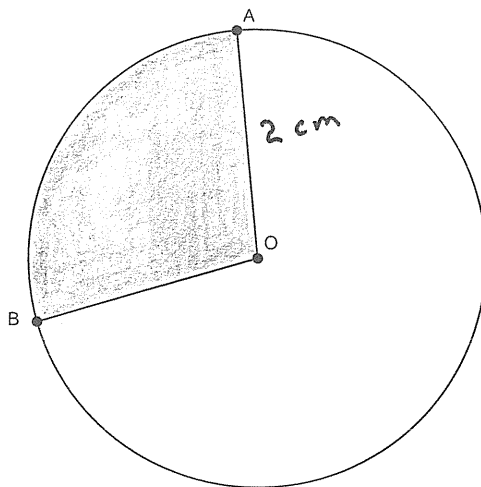
a) Find $\lim_{x \rightarrow 0} \frac{\sin\left(\frac{2x}{5}\right)}{3x}$ 2

b) (i) Show that $\cos 4x = 2\cos^2 2x - 1$ 1

(ii) Hence find $\int \cos^2 2x \, dx$ 2

c) Evaluate $\int_0^{\log_e 2} \frac{e^{3x}}{1+e^{3x}} \, dx$ 3

d)



The area of the minor sector AOB is 5 cm^2 .

(i) Find the size of $\angle AOB$ 2

(ii) Hence, find the length of the major arc AB (correct to 1 decimal place) 2

Question 3 – (12 marks) – (Start a new booklet)

Marks

- a) (i) Show that the line $y = 2 + 2x$ crosses the curve $y = \ln(3x + 4)$ at the point $(-1, 0)$

1

- (ii) Find the acute angle between $y = 2 + 2x$ and $y = \ln(3x + 4)$ at the point $(-1, 0)$. [answer to nearest minute]

3

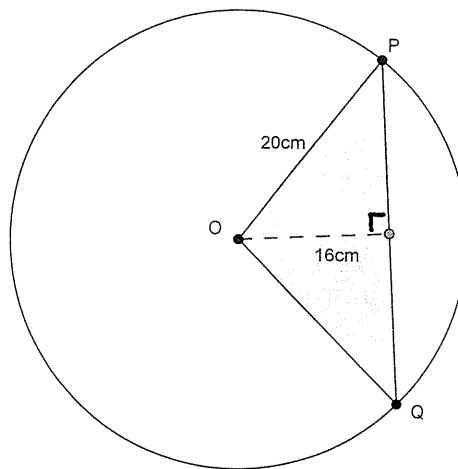
- b) Evaluate

2

$$\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \tan^2 x \, dx$$

- c) Calculate the area of the minor segment cut off by the chord PQ , given the perpendicular distance from the centre O to PQ is 16cm.

3



- d) Draw a neat sketch of the curve $y = \ln(3 - x)$ clearly indicating any x, y intercepts and asymptote(s).

3

Question 4 – (12 marks) – (Start a new booklet)

Marks

- a) (i) Find $\frac{d}{dx} \left[\frac{\log_e x}{x^2} \right]$ 2
- (ii) Hence, (or otherwise) find the equation of the tangent to the curve
 $y = \frac{\log_e x}{x^2}$, at $x = e$ 2
- b) The area between the curves $y = \cos x$ and $y = \sin x$ from $x = 0$ to $x = \frac{\pi}{4}$ is rotated about the x -axis. 3
- Calculate the volume of the solid formed. [give in exact form]
- c) For the curve $y = e^{\sin x} - 1$ in the domain $0 \leq x \leq 2\pi$ find:
- (i) the x -intercepts 2
- (ii) the stationary points and determine their nature. 3

Question 5 – (12 marks) – (Start a new booklet)

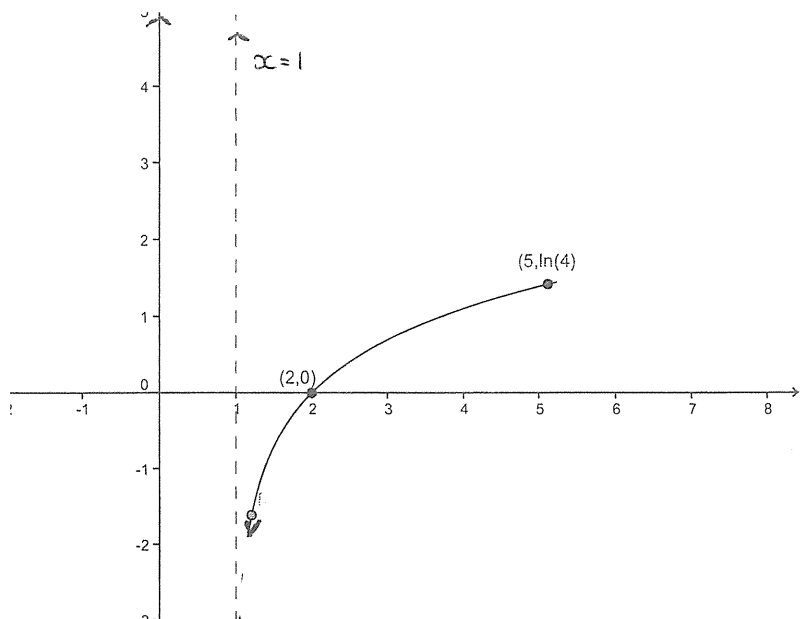
Marks

- a) (i) Sketch the graph of $y = \sin 2x$ in the domain $0 \leq x \leq 2\pi$
[at least $\frac{1}{3}$ page] 2
- (ii) On the same graph, sketch the line required to determine the number
of solutions for the equation $\frac{x}{2} + \sin 2x = 1$ 2
- (iii) How many solutions are there? 1
- b) Find: $\int \cos^3 x \cdot \sin x \, dx$ 2
- c) (i) Prove $\tan(45^\circ - A) = \frac{1 - \sin 2A}{\cos 2A}$ where A is acute. 3
- (ii) Hence, using a suitable value for A , find the exact value for $\tan 15^\circ$ 2

Question 6 – (12 marks) – (Start a new booklet)

Marks

- a) A sketch of $y = \ln(x - 1)$ for $1 < x \leq 5$ is given.



- (i) Show that the point $(e + 1, 1)$ lies on the curve. 1
- (ii) Calculate the exact area bound by the curve $y = \ln(x - 1)$, the x -axis and the line $x = e + 1$ 3
- b) (i) Differentiate $x \log_e x - x$ 1
- (ii) Hence, evaluate $\int_1^e \log_e x \, dx$ 3
- c) If $y = e^x \cdot \cos x$, find the value of k for which

$$\frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} + ky = 0$$

is true for all x .

4

Question 1

YEAR 12 EXT 1 - MIDCOURSE HSC SOLNS

a) $\log_e 5 - \frac{1}{2} \log_e 16 = \log_e k$

LHS $\log_e 5 - \log_e 16^{\frac{1}{2}}$

$$= \log_e 5 - \log_e 4$$

$$= \log_e \left(\frac{5}{4}\right)$$

$$\therefore \underline{\underline{k = \frac{5}{4}}}$$

b) (i) $\frac{2\pi}{9}$ radians

$$\pi \text{ radians} = 180^\circ$$

$$\therefore \frac{2\pi}{9} \text{ radians} = 180^\circ \times \frac{2}{9}$$

$$= 40^\circ$$

(ii) $\cos\left(-\frac{11\pi}{6}\right) + \sin\left(\frac{2\pi}{3}\right)$

$$= \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2}$$

$$= \frac{2\sqrt{3}}{2}$$

$$= \sqrt{3}$$

c) $y = \log_e \left(\frac{x+1}{\sqrt{x}}\right)$

$$y = \log_e (x+1) - \log_e \sqrt{x}$$

$$y = \log_e (x+1) - \log_e x^{\frac{1}{2}}$$

$$y = \log_e (x+1) - \frac{1}{2} \log_e x$$

then

$$\frac{dy}{dx} = \frac{1}{x+1} - \frac{1}{2} \times \frac{1}{x}$$

$$= \frac{1}{x+1} - \frac{1}{2x}$$

$$= \frac{2x - (x+1)}{2x(x+1)}$$

$$= \frac{2x - x - 1}{2x(x+1)}$$

$$= \frac{x-1}{2x(x+1)}$$

d) $y = \log_e (x^2 + 2)$ where $x=1$.

$$\frac{dy}{dx} = \frac{2x}{x^2+2}$$

When $x=1$, $\frac{dy}{dx} = \frac{2}{1^2+2}$

$$\frac{dy}{dx} = \frac{2}{3}$$

$$\therefore m = \frac{2}{3}$$

∴ gradient of normal = $-\frac{3}{2}$

For equation, $x_1=1, y_1 = \log_e 3$
 $y - y_1 = m(x - x_1)$

$$y - \log_e 3 = -\frac{3}{2}(x - 1)$$

$$y - \log_e 3 = -\frac{3}{2}x + \frac{3}{2}$$

$$2y - 2\log_e 3 = -3x + 3$$

$$3x + 2y = 3 + 2\log_e 3$$

For coordinates,

when $x=1$, $y = \log_e (1^2 + 2)$

$$y = \log_e 3$$

$$(1, \log_e 3)$$

Qn 2

a) $\lim_{x \rightarrow 0} \frac{\sin(\frac{2x}{5})}{3x}$

$$= \frac{1}{3} \lim_{x \rightarrow 0} \frac{\frac{2}{5} \sin(\frac{2x}{5})}{(\frac{2x}{5})}$$

$$= \frac{2}{15} \lim_{x \rightarrow 0} \frac{\sin(\frac{2x}{5})}{(\frac{2x}{5})}$$

$$= \frac{2}{15}$$

b) (i) Show that $\cos 4x = 2\cos^2 2x - 1$

$$\begin{aligned} \text{LHS } \cos(2x+2x) &= \cos 2x \cos 2x - \sin 2x \sin 2x \\ &= \cos^2 2x - \sin^2 2x \\ &= \cos^2 2x - (1 - \cos^2 2x) \\ &= \cos^2 2x - 1 + \cos^2 2x \\ &= 2\cos^2 2x - 1 \\ &= \text{RHS} \end{aligned}$$

(ii) $\int \cos^2 2x \, dx = \int \frac{\cos 4x}{2} + \frac{1}{2} \, dx$

$$= \frac{1}{2} \times \frac{1}{4} \sin 4x + \frac{x}{2} + C$$

$$= \frac{\sin 4x}{8} + \frac{x}{2} + C$$

$$\cos 4x = 2\cos^2 2x - 1$$

$$\frac{1}{2} \cos 4x = \cos^2 2x - 1$$

$$\therefore \frac{1}{2} \cos 4x + 1 = \cos^2 2x$$

$$\cos^2 2x = \frac{\cos 4x}{2} + 1$$

$$c) \int_0^{\log_e 2} \frac{e^{3x}}{1+e^{3x}} dx$$

$$= \frac{1}{3} \int_0^{\log_e 2} \frac{3e^{3x}}{1+e^{3x}} dx$$

$$= \frac{1}{3} \left[\log_e (1+e^{3x}) \right]_0^{\log_e 2}$$

$$= \frac{1}{3} \left[\log_e (1+e^{3 \log_e 2}) - \log_e (1+e^0) \right]$$

$$= \frac{1}{3} \left[\log_e (1+e^{\log_e 2^3}) - \log_e 2 \right]$$

$$= \frac{1}{3} \left[\log_e (1+2^3) - \log_e 2 \right]$$

$$= \frac{1}{3} \left[\log_3 9 - \log_e 2 \right]$$

$$= \frac{1}{3} \log_e \left(\frac{9}{2} \right)$$

$$d) \text{ (i) Area of minor sector} = 5 \text{ cm}^2$$

$$\frac{1}{2} r^2 \theta = 5$$

$$\frac{1}{2} \times 2^2 \theta = 5$$

$$2\theta = 5$$

$$\therefore \theta = \frac{5}{2} \text{ radians.}$$

$$\text{(ii) Length of major arc AB.}$$

$$l = r\theta. \quad \theta = 2\pi - \frac{5}{2}$$

$$l = 2 \times \left(\frac{4\pi - 5}{2} \right) \quad \theta = \frac{4\pi - 5}{2}$$

$$l = 4\pi - 5 \quad \theta = \frac{4\pi - 5}{2}$$

$$l = 7.566370614$$

$$\therefore l = \underline{7.6} \text{ cm. (to 1 dec place).}$$

Qn3

$$a \text{ (i)} \quad y = 2 + 2x \quad y = \ln(3x+4)$$

When $x = -1$,

$$y = 2 + 2(-1)$$

$$y = 2 - 2$$

$$y = 0.$$

$$y = \ln(3(-1)+4)$$

$$y = \ln(-3+4)$$

$$y = \ln 1$$

$$y = 0$$

$\therefore$ The line and the curve cross at the point $(-1, 0)$.

$$c(i) \quad \tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

For $y = 2 + 2x$

For $y = \ln(3x+4)$

$$\frac{dy}{dx} = 2$$

$$\frac{dy}{dx} = \frac{3}{3x+4}$$

Let $\underline{m_1 = 2}$

When $x = -1$, $\frac{dy}{dx} = \frac{3}{-3+4}$

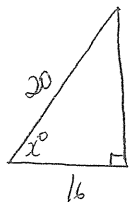
Let $\underline{m_2 = 3}$.

$$\text{Then } \tan \theta = \left| \frac{2-3}{1+2 \times 3} \right| \quad \tan \theta = \left| \frac{-1}{7} \right|$$

$$\therefore \underline{\theta = 8^\circ 8'}$$

$$\begin{aligned} b) \quad \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \tan^2 x \, dx &= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \sec^2 x - 1 \, dx \\ &= \left[\tan x - x \right]_{\frac{\pi}{4}}^{\frac{\pi}{3}} \\ &= \left[\left(\tan \frac{\pi}{3} - \frac{\pi}{3} \right) - \left(\tan \frac{\pi}{4} - \frac{\pi}{4} \right) \right] \\ &= \sqrt{3} - \frac{\pi}{3} - \left(1 - \frac{\pi}{4} \right) \\ &= \sqrt{3} - \frac{\pi}{3} - 1 + \frac{\pi}{4} \\ &= \underline{\underline{\sqrt{3} - 1 - \frac{\pi}{12}}} \end{aligned}$$

c) Area of Minor Segment = $\frac{1}{2}r^2(\theta - \sin\theta)$



$$= \frac{1}{2} \times 20^2 (\theta - \sin\theta)$$

$$= \frac{1}{2} \times 20^2 (1.286... - \sin 1.286...)$$

$$= 65 \text{ cm}^2$$

$$\cos x^\circ = \frac{16}{20}$$

$$x = 36^\circ 52'$$

OR Let $\hat{POQ} = 2\alpha$
 $\cos \alpha = \frac{16}{20}$
 $= \frac{4}{5}$

Now π radians = 180°

$$\frac{\pi}{180} \text{ radians} = 1^\circ$$

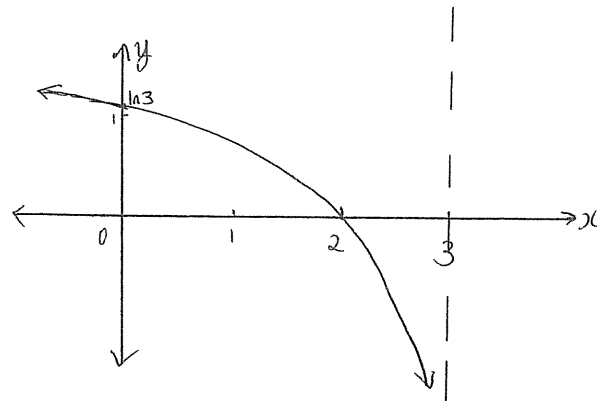
$$\frac{36.8 \cdot \pi}{180} \text{ radians} = 36^\circ 52'$$

We need $2x = 73^\circ 44'$

$$\frac{73^\circ 44' \pi}{180} \text{ radians} = 73^\circ 44'$$

$$1.286889435 \text{ radians} = 73^\circ 44'$$

d) $y = \ln(3-x)$



when $x=0$,
 $y = \ln 3$

Area minor segment

$$= \frac{1}{2}r^2(\theta - \sin\theta)$$

$$= \frac{1}{2}r^2[2\alpha - \sin 2\alpha]$$

$$= 200 \left[2\cos^{-1}\left(\frac{4}{5}\right) - 2 \cdot \frac{3}{5} \cdot \frac{4}{5} \right]$$

$$= 65.400...$$

$$= 65.4 \text{ cm}^2 \text{ (correct to 1 dec pl)}$$

Qn 4

a (i) $\frac{d}{dx} \left[\frac{\log_e x}{x^2} \right]$

$$= \frac{x^2 \cdot \frac{1}{x} - \log_e x \cdot 2x}{(x^2)^2}$$

$$= \frac{x - 2x \log_e x}{x^4}$$

$$= \frac{1 - 2 \log_e x}{x^3}$$

ii) $\frac{dy}{dx} = \frac{1 - 2 \log_e x}{x^3}$

when $x=e$,

$$\frac{dy}{dx} = \frac{1 - 2 \log_e e}{e^3}$$

$$m = \frac{1-2}{e^3}$$

$$m = \frac{-1}{e^3}$$

For coordinates, when $x=e$,

$$y = \frac{\log_e e}{e^2}$$

$$y = \frac{1}{e^2}$$

$$\therefore (e, \frac{1}{e^2})$$

Eqn of tangent

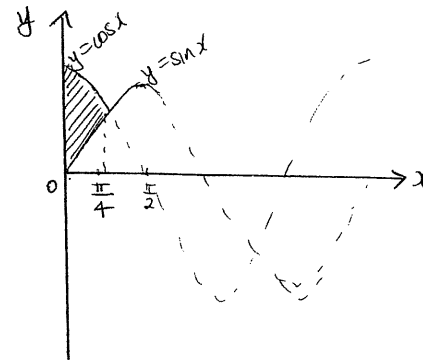
$$y - \frac{1}{e^2} = -\frac{1}{e^3} (x - e)$$

$$e^3 y - e = -x + e$$

b)

$$y = \cos x$$

$$y = \sin x$$



$$V = \pi \int_0^{\pi/4} \cos^2 x - \sin^2 x \, dx$$

$$V = \pi \int_0^{\pi/4} \cos 2x \, dx$$

$$V = \pi \left[\frac{1}{2} \sin 2x \right]_0^{\pi/4}$$

$$V = \frac{\pi}{2} \left[\sin 2x \right]_0^{\pi/4}$$

$$V = \frac{\pi}{2} \left[\sin \frac{\pi}{2} - \sin 0 \right]$$

$$V = \frac{\pi}{2} \left[\sin \frac{\pi}{2} - \sin 0 \right]$$

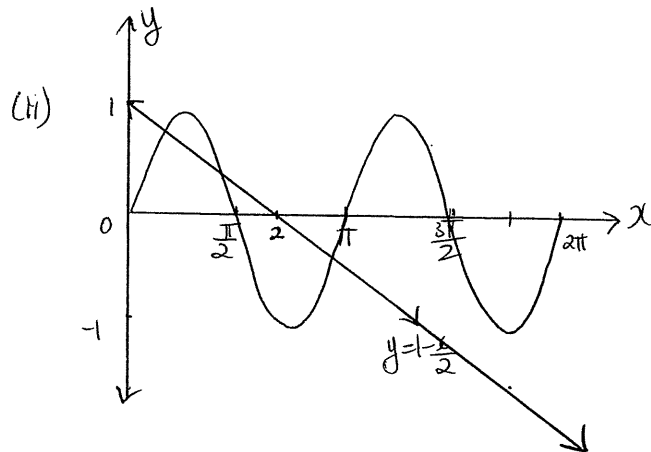
$$V = \frac{\pi}{2} [1] \quad \therefore V = \frac{\pi}{2} \text{ units}^3$$

Qn5

a) (i) $y = \sin 2x$ for $0 \leq x \leq 2\pi$

$a=1$, Period $= \frac{2\pi}{2}$

Period $= \pi$ units.



$$\frac{x}{2} + \sin 2x = 1$$

$$\sin 2x = 1 - \frac{x}{2}$$

For $y = 1 - \frac{x}{2}$

When $x=0$, $y=1$

$y=0$, $x=2$

(iii) There are 3 solutions.

b) $\int \cos^3 x \sin x dx$

$$= \int \sin x \cdot (\cos x)^3 dx$$

$$= - \int -\sin x (\cos x)^3 dx$$

$$= - \left[\frac{(\cos x)^4}{4} \right] + C$$

$$= - \frac{\cos^4 x}{4} + C$$

c) (i) Prove $\tan(45^\circ - A) = \frac{1 - \sin 2A}{\cos 2A}$ where A is acute

LHS. $\tan(45^\circ - A) = \frac{\tan 45^\circ - \tan A}{1 + \tan 45^\circ \tan A}$

$$= \frac{1 - \tan A}{1 + \tan A}$$

$$= \frac{1 - \frac{\sin A}{\cos A}}{1 + \frac{\sin A}{\cos A}} \times \cos A$$

$$= \frac{\cos A - \sin A}{\cos A + \sin A}$$

c) $y = e^{\sin x} - 1 \quad 0 \leq x \leq 2\pi$

(i) x -intercepts when $y=0$

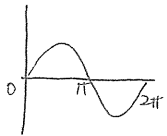
$$e^{\sin x} - 1 = 0$$

$$e^{\sin x} = 1$$

$$\ln e^{\sin x} = \ln 1$$

$$\sin x = 0$$

$$\therefore x = 0, \pi, 2\pi$$



(ii) Possible stationary pts occur when $\frac{dy}{dx} = 0$.

$$\text{ie } \cos x e^{\sin x} = 0$$

$$\text{ie } \cos x = 0 \text{ or } e^{\sin x} \neq 0 \text{ no solns}$$

$$\therefore x = \frac{\pi}{2}, \frac{3\pi}{2}$$

OR Test $\frac{dy}{dx}$ either side.

To test nature.

$$\begin{aligned} \frac{dy}{dx} &= e^{\sin x} (-\sin x) + \cos x \cdot \cos x e^{\sin x} \\ &= -\sin x e^{\sin x} + \cos^2 x e^{\sin x} \end{aligned}$$

$$\frac{dy}{dx} = e^{\sin x} (\cos^2 x - \sin x)$$

$$\text{when } x = \frac{\pi}{2}, \quad \frac{dy}{dx} = e^{\sin \frac{\pi}{2}} \left(\cos^2 \frac{\pi}{2} - \sin \frac{\pi}{2} \right)$$

$$= e(0 - 1)$$

$$= -e$$

$$\frac{dy}{dx} < 0 \quad \therefore \text{max when } x = \frac{\pi}{2}$$

For coordinates, when $x = \frac{\pi}{2}$, $y = e^{\sin \frac{\pi}{2}} - 1$ Local maximum at $(\frac{\pi}{2}, e-1)$
 $y = e - 1$

$$\text{when } x = \frac{3\pi}{2}, \quad \frac{dy}{dx} = e^{\sin \frac{3\pi}{2}} \left(\cos^2 \frac{3\pi}{2} - \sin \frac{3\pi}{2} \right)$$

$$= e^{-1}(0 - -1)$$

$$= \frac{1}{e} > 0$$

$\therefore$ minimum turning pt.

For coordinates, when $x = \frac{3\pi}{2}$, $y = e^{\sin \frac{3\pi}{2}} - 1$ Local minimum at $(\frac{3\pi}{2}, \frac{1}{e} - 1)$
 $y = e^{-1} - 1$

$$= \frac{\cos A - \sin A}{\cos A + \sin A} \times \frac{\cos A - \sin A}{\cos A - \sin A}$$

$$= \frac{\cancel{\cos^2 A} - 2\sin A \cos A + \cancel{\sin^2 A}}{\cos^2 A - \sin^2 A}$$

$$= \frac{1 - 2\sin A \cos A}{\cos^2 A - \sin^2 A}$$

$$= \frac{1 - \sin 2A}{\cos 2A}$$

$$= \text{RHS}$$

Since
 $\sin 2A = 2\sin A \cos A$
 $\cos 2A = \cos^2 A - \sin^2 A$

(11) $\tan 15^\circ$

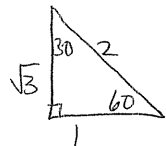
$$= \tan(45^\circ - 30^\circ)$$

$$= \frac{1 - \sin(2 \times 30^\circ)}{\cos(2 \times 30^\circ)}$$

$$= \frac{1 - \sin 60^\circ}{\cos 60^\circ}$$

$$= \frac{1 - \frac{\sqrt{3}}{2}}{\frac{1}{2}}$$

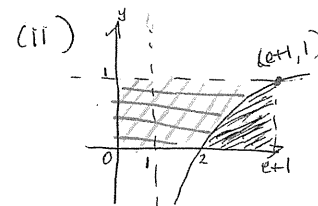
$$= \frac{2 - \sqrt{3}}{\frac{1}{2}} = 2 - \sqrt{3} \text{ units.}$$



Question 6

a) (i) when $x = e+1$, $y = \ln(e+1-1)$
 $y = \ln e$
 $y = 1$

$\therefore (e+1, 1)$ lies on the curve.



$$\text{Area} = \text{Area of rectangle} - \int_0^1 e^{y+1} dy.$$

$y = \ln(x-1)$
 $x-1 = e^y$
 $x = e^y + 1$

$$= (e+1) - [e^y + y]_0^1$$

$$= e+1 - [(e+1) - (e^0 + 0)]$$

$$= e+1 - [e+1-1]$$

$$= 1 \text{ unit}^2$$

or: $\frac{d}{dx} (x-1) \ln(x-1)$
 $= \ln(x-1) + 1$

$$A = \int_2^{e+1} \ln(x-1) dx = \int_2^{e+1} (\ln(x-1) + 1) dx - \int_2^{e+1} 1 dx$$

$$= [(x-1) \ln(x-1) - x]_2^{e+1}$$

$$= (e \ln e - e - 1) - (\ln 1 - 2)$$

$$= e - e - 1 - 0 + 2 = 1 \text{ u}^2$$

$$(b) (i) x \log_e x - x$$

Differentiating,

$$\log_e x \cdot 1 + x \cdot \frac{1}{x} - 1$$

$$= \log_e x + 1 - 1$$

$$= \log_e x.$$

$$\therefore \frac{d}{dx} (x \log_e x - x) = \log_e x$$

$$(ii) \int_1^e \log_e x \, dx = \int_1^e \frac{d}{dx} (x \log_e x - x) \, dx$$

$$= [x \log_e x - x]_1^e$$

$$= [(e \log_e e - e) - (1 \log_e 1 - 1)]$$

$$= [(e - e) - (0 - 1)]$$

$$= \underline{1}$$

$$c) y = e^x \cdot \cos x$$

$$\frac{dy}{dx} = \cos x \cdot e^x + e^x \cdot -\sin x$$

$$= e^x (\cos x - \sin x)$$

$$\frac{d^2y}{dx^2} = (\cos x - \sin x) \cdot e^x + e^x (-\sin x - \cos x)$$

$$= e^x (\cos x - \sin x) - e^x (\sin x + \cos x)$$

$$= e^x \cancel{\cos x} - e^x \sin x - e^x \sin x - e^x \cancel{\cos x}$$

$$\frac{d^2y}{dx^2} = -2e^x \sin x$$

Substituting,

$$-2e^x \sin x - 2e^x (\cos x - \sin x) + ke^x \cos x = 0$$

$$-2e^x \cancel{\sin x} - 2\cos x e^x + 2e^x \cancel{\sin x} + ke^x \cos x = 0$$

$$-2\cos x e^x + ke^x \cos x = 0$$

$$-2e^x \cos x + ke^x \cos x = 0$$

$$e^x \cos x (-2 + k) = 0$$

$$e^x \cos x (k - 2) = 0$$

$$\therefore \underline{k = 2}$$